VECTOR

MICROSAR VTTMemMap

Technical Reference

VTT Memory Mapping Generator

Version 1.0.0

Authors	virleh
Status	Released

Document Information

History

Author	Date	Version	Remarks
virleh	2022-03-09	1.0.0	Initial version

Reference Documents

No.	Source	Title	Version
[1]	AUTOSAR	List of Basic Software Modules	R19-11
[2]	Vector	MICROSAR MemMap Technical Reference	See delivery information



Caution
We have configured the programs in accordance with your specifications in the questionnaire. Whereas the programs do support other configurations than the one specified in your questionnaire, Vector´s release of the programs delivered to your company is expressly restricted to the configuration you have specified in the questionnaire.

Contents

1 Introduction 5

2 Functional Description 6
2.1 Initialization 6
2.2 Main Functions 6
2.3 Automatic Pragma Generation 6
2.3.1 Default behavior 6
2.3.2 AUTOSAR Compliance 6

3 Integration 7
3.1 Embedded Implementation 7
3.2 Static Code Analysis 7

4 API Description 8

5 Configuration 9
5.1 Configuration Variants 9

6 Glossary and Abbreviations 10
6.1 Glossary 10
6.2 Abbreviations 10

7 Contact 11

Tables

Table 3-1 Implementation files 7
Table 6-1 Glossary 10
Table 6-2 Abbreviations 10

1 Introduction

This document describes the functionality, API and configuration of the BSW module VTTMemMap.

Supported Configuration Variants:	pre-compile	
Vendor ID:	VTTMemMap_VENDOR_ID	30 decimal (= Vector-Informatik, according to HIS)
Module ID:	VTTMemMap_MODULE_ID	195 decimal (according to ref. [1])

The VTTMemMap is the implementation of the MICROSAR MemMap for vVIRTUALtarget (VTT). The behavior of the VTTMemMap is like the MICROSAR MemMap as described in [2]. For this reason, general information is available in the description of the MICROSAR MemMap. This document describes specific details of the VTTMemMap.

2 Functional Description

2.1 Initialization

The VTTMemMap only generates memory mapping header files, thus initialization is not applicable.

2.2 Main Functions

The VTTMemMap only generates memory mapping header files, thus main functions are not applicable.

2.3 Automatic Pragma Generation

The VTTMemMap does not support automatic pragma generation as described in [2]. Therefore, setting a reference to a vLinkGen logical group has no effect. The VTTMemMap does generate memory allocation keywords, however these do not contain any pragmas by default.

If pragmas are needed, it is possible to enter them manually (see 2.3.1).

2.3.1 Default behavior

If no manually entered pragmas are present, the behavior depends on the compiler used:

- When using the MinGW compiler, usage of memory allocation keywords related to variable data results in a mapping to the statically defined linker section `vttdata`. The memory allocation keywords related to code and constant data will not lead to any mapping. Because of that their usage has no effect
- If another compiler is used the usage of memory allocation keywords will not lead to any mapping by default. Because of that their usage has no effect

2.3.2 AUTOSAR Compliance

It is possible to specify pragma statements directly in the configuration and generate them into the memory allocation keywords, as it is specified by AUTOSAR. To do so, the following parameters can be used to specify the pragma statements to start or to stop a memory section:

- `/MICROSAR/VTT/VTTMemMap/MemMapAddressingModeSet/MemMapAddressingMode/MemMapAddressingModeStart`
- `/MICROSAR/VTT/VTTMemMap/MemMapAddressingModeSet/MemMapAddressingMode/MemMapAddressingModeStop`

3 Integration

This chapter gives necessary information for the integration of the MICROSAR VTTMemMap into an application environment of an ECU.

3.1 Embedded Implementation

The delivery of the VTTMemMap contains these source code files:

File Name	Description	Integration Task
<Mip>_MemMap.h	Generated memory mapping header file for a dedicated BSW module.	-
<Ctn>_MemMap.h	Generated memory mapping header file for a dedicated software component type.	-
MemMap_Compatibility.h	Generated memory mapping header file for compatibility with non AUTOSAR 4.x compliant BSW modules. Contains all memory allocation keywords of all BSW modules from the input configuration.	
MemMap_Common.h	Generated memory mapping header file for compatibility with non AUTOSAR MemMap 4.x compliant BSW modules. Contains standard memory allocation keywords for BSW modules not using the AUTOSAR MemMap 4.x mechanism. For example: `START_SEC_CODE` and `STOP_SEC_CODE`.	-

Table 3-1 Implementation files

3.2 Static Code Analysis

When running static code analysis tools, the preprocessor definition `MICROSAR_DISABLE_MEMMAP` should be set. It disables the memory mapping header files via the preprocessor and thus also the use of compiler specific pragmas. This helps to avoid errors, since static code analysis tools often cannot handle pragmas.

4 API Description

The VTTMemMap only generates memory mapping header files, thus API Description is not applicable.

© 2022 Vector Informatik GmbH

5 Configuration

MICROSAR VTTMemMap is configured with Vectors DaVinci Configurator.

Additional information about all configuration attributes can be found within the configuration tool.

5.1 Configuration Variants

The VTTMemMap supports the configuration variants

> `VARIANT-PRE-COMPILE`

The configuration classes of the VTTMemMap parameters depend on the supported configuration variants. For their definitions please see the *MemMap_bswmd.arxml* file.

6 Glossary and Abbreviations

6.1 Glossary

Term	Description
BSWMD	The BSWMD is a formal notation of all information belonging to a certain BSW artifact (BSW module or BSW cluster) in addition to the implementation of that artifact.
Electronic Control Unit	Also known as ECU. Small embedded computer system consisting of at least one CPU and corresponding periphery which is placed in one housing.
Memory Mapping Header File	Header file generated by the MemMap containing the memory allocation keywords of all assigned memory sections.
Memory Allocation Keyword	Refer to see [2]
Memory Section	Refer to see [2]
SwAddrMethod	Refer to see [2]
Addressing Mode Set	Refer to see [2]
Allocation	Refer to see [2]

Table 6-1 Glossary

6.2 Abbreviations

Abbreviation	Description
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
BSW	Basic Software
CTN	Component Type Name (see [1]).
DEM	Diagnostic Event Manager
DET	Development Error Tracer
ECU	Electronic Control Unit
HIS	Hersteller Initiative Software
MemMap	Memory Mapping
MICROSAR	Microcontroller Open System Architecture (the Vector AUTOSAR solution)
MIP	Module Integration Package which is composed according `<Msn>[_<vi>_<ai>]` for BSW modules (see [1]).
SWC	Software Component
SWS	Software Specification
vLinkGen	Vector’s Compiler Specific Linker Script Generator

Table 6-2 Abbreviations

7 Contact

Visit our website for more information on

- News
- Products
- Demo software
- Support
- Training data
- Addresses

www.vector.com

© 2022 Vector Informatik GmbH